3.3 Linear Functions

Algebra I 3.3 · Student sheet

Name _____ Period ____ Date _____

I can…

- tell from a graph or a table whether a function is linear.
- tell from an equation whether a function is linear.
- decide whether a domain is discrete or continuous.

Vocabulary · fill in the missing word

Linear function — a function whose graph is a _____ line; it has a constant rate of change

Nonlinear function — a function without a _____ rate of change

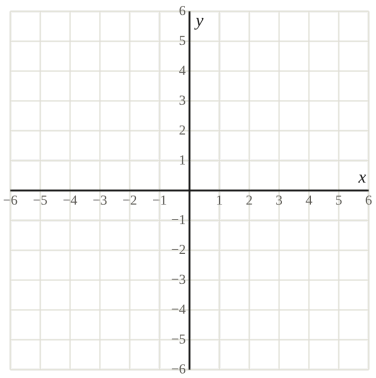
Discrete domain — _____ input values (counts, such as people or tickets)

Continuous domain — all values in an _____ (measurements, such as time or weight)

Part 1 · Linear or not?

x	0	1	2	3	4	Changes in y	Linear?
Table A: y	2	5	8	11	14		
Table B: y	1	2	4	8	16		
Table C: y	0	1	4	9	16		
Table D: y	10	8	6	4	2		

Graph two of the tables:



Part 2 · Notes and examples (fill in each note, then work the example)

Linear functions: A function is linear when its rate of change is _____.

Example 1

Is each table linear?

a. x: 0, 2, 4, 6; y: 1, 5, 9, 13

b. x: 1, 2, 3, 4; y: 1, 4, 9, 16

Nonlinear: $y = x^2$ and $y = 5/x$ are _____.

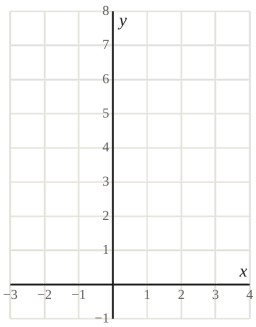
Example 2

Is each equation linear?

a. $y = 3x - 7$ b. $y = x^2 + 1$ c. $y = \frac{5}{x}$ d. $2x + 3y = 6$

Example 3

The graph shows two functions. Which one is linear?



Discrete and continuous: Discrete domain: you can only _____. Continuous domain: you can _____.

Example 4

Is the domain discrete or continuous?

a. the number of tickets sold and the money collected

b. the gallons of gas pumped and the cost

Part 3 · Partner practice

Partner 1

Is each equation linear? a. $y = 4 - 2x$ b. $y = x^3$

Partner 2

Is the domain discrete or continuous? a. a puppy's weight over time b. the number of people in a car